

Rishav Singh

Data Scientist / ML Engineer / AI Engineer

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Summary

I'm Rishav Singh, **AI & Data Science undergraduate** with a strong interest in **applied machine learning** and **generative AI**. Hands-on experience building projects involving **data analysis**, **ML modelling**, and **LLM-based applications** using **Python**. Familiar with **RAG pipelines**, **prompt engineering**, and integrating **AI models through APIs**. Actively developing skills in **data-driven problem solving**, **model evaluation**, and **practical AI system design**.

Projects

- **NyayaAI – AI-Powered Legal Assistant (LLM, RAG, LangChain, NLP, Flask, Web Search)**
Built a full-stack **AI-powered legal assistant** enabling context-aware legal question answering through **Retrieval-Augmented Generation (RAG)** and intelligent web search. Designed a stateless **Flask API backend** orchestrating **LLM reasoning**, **document-based contextual chat**, and **search-augmented responses** using Gemini and LangChain. Implemented **document upload**, **session-based memory handling**, and modular API endpoints with logging and error handling.
Tech stack: Python, Flask, LangChain, Gemini API, Web Search, HTML, Tailwind CSS, JavaScript.
- **ResolvAI – Offline AI Complaint Resolution System (Local LLMs, GenAI, LangChain, Streamlit)**
Developed an **offline, privacy-preserving AI application** to generate professional and empathetic responses to customer complaints for small businesses. Integrated **local LLMs via Ollama** with **LangChain** to analyze complaint text, business profiles, and policy documents for **context-aware response generation**. Implemented **session-level conversation history**, tone customization, and a **Streamlit-based interactive UI**, ensuring complete offline operation and data security.
Tech stack: Python, LangChain, Ollama, Streamlit, SQLite, Local LLMs (Llama3, Mistral).
- **VedaTale – AI-Generated Storytelling Platform (LLM, Flask, Web Development, Firebase)**
Built an **AI-powered storytelling web platform** that generates narrative content from user prompts and enables users to explore and share stories. Developed a **Flask backend** for AI-driven story generation and a multi-page **frontend interface** with galleries and user interaction features. Deployed the static frontend using **Firebase Hosting** and structured the backend for scalable cloud deployment.
Tech stack: Python, Flask, LLM APIs, HTML, CSS, JavaScript, Firebase Hosting.
- **Tulsi Disease Detection – ML API with Frontend (Computer Vision, CNNs, TensorFlow, OpenCV)**
Developed an **end-to-end computer vision system** for plant disease detection using **custom CNN models** trained on labelled image datasets. Implemented **image preprocessing**, **model training**, and **performance comparison** through automated training and analysis scripts. Built a **REST API** for real-time inference and integrated a browser-based **frontend** to display predictions, confidence scores, and treatment recommendations.
Tech stack: Python, TensorFlow/Keras, OpenCV, CNNs, REST API, HTML, CSS.

Professional Skills

Core Domains: Machine Learning, Data Science, Generative AI, Natural Language Processing (NLP), Computer Vision, LLM-based Systems, Retrieval-Augmented Generation (RAG)

Programming Languages: Python, SQL, HTML, CSS, JavaScript

ML & AI Frameworks: scikit-learn, TensorFlow/Keras, PyTorch, LangChain

LLMs & GenAI Tools: Prompt Engineering, Embeddings, Local LLMs, LLM API Integration, Context Management

Data & CV Libraries: NumPy, pandas, OpenCV, Matplotlib, NLTK / spaCy (basic)

Backend & APIs: Flask, REST API Development, Model Inference Pipelines

Databases & Storage: SQLite, Vector Databases (conceptual understanding)

Deployment & Tools: Git, GitHub, Docker (basic), Jupyter Notebooks, Firebase Hosting

Foundations: Statistics & Probability, Linear Algebra, Model Evaluation, Data Preprocessing, Feature Engineering

Education

Bachelor of Technology (B. Tech) - Computer Science Engineering (AI & Data Science)

MIT World Peace University (MIT-WPU), Pune

CGPA: 7.84 | Duration: Aug 2023 – Present

Senior Secondary (Class XII) – CBSE

Shree Swaminarayan Gurukul, Salvav, Vapi

Percentage: 64.8% | 2022 – 2023

Secondary (Class X) – CBSE

Shree Swaminarayan Gurukul, Salvav, Vapi

Percentage: 79.8% | 2020 – 2021

Certificates & Courses

- **IBM Professional Data Scientist Certificate** – IBM
- **Advanced Deep Learning** – iNeuron.AI
- **Python & Web Development** – Certificate Course

Soft Skills

- **Analytical Thinking:** Applied analytical reasoning to data analysis, model evaluation, and understanding complex AI/ML systems across academic and personal projects.
- **Problem Solving:** Experienced in interpreting data, implementing algorithms, and iterating on solutions to address technical challenges in ML and AI applications.
- **Collaboration:** Worked effectively in team-based academic and project environments, coordinating requirements and contributing to shared technical goals.
- **Communication:** Able to explain AI/ML concepts clearly through technical writing, presentations, and discussions with both technical and non-technical audiences.
- **Continuous Learning:** Actively learning new tools, frameworks, and concepts in machine learning, generative AI, and software development.
- **Adaptability:** Quick to learn and apply new technologies, frameworks, and workflows as project requirements evolve.
- **Time Management:** Capable of managing multiple projects, coursework, and deadlines while maintaining consistent progress and quality.